

Problem-Solution Fit

Date	10/07/2026
Name	Jaswanth Madapuram
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Banks • Financial Institutions • Credit Card Applicants • Loan Officers / Credit Analysts	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited or no credit history • Incomplete or incorrect application information • Low credit score • Income instability • Budget constraints • Lack of digital / technical knowledge	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Verification - Pros: Accurate - Cons: Slow, time consuming, prone to human error • Rule-Based Systems - Pros: Easy to implement, fast decision - Cons: Inflexible, limited accuracy • ML/AI Based Models - Pros: High accuracy, fast, scalable, learns from data - Cons: Requires quality data and model maintenance
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual approval process is slow (Very frequent) • High risk of fraudulent approvals (Very frequent) • Incorrect approvals or rejections (Frequent) • Time-consuming document verification (Very frequent) • Human errors in decision making (Frequent) • High operational cost (Frequent)	9. PROBLEM ROOT / CAUSE [RC] • Manual processing and verification • Incomplete or inaccurate applicant information • Inconsistent decision making • Lack of predictive analysis • Fraudulent / identity theft cases • No real-time risk assessment	7. BEHAVIOR + ITS INTENSITY [BE] • Expect quick approval (High intensity) • Apply online frequently (High intensity) • Submit required documents when requested (Medium intensity) • Compare offers from multiple banks before applying (High intensity)
3. TRIGGERS TO ACT [TR] • Customer submits a credit card application • Need for faster approval • Increase in application volume • Rising fraud cases • Business need to reduce default risk	10. YOUR SOLUTION [SL] Develop a Machine Learning based Credit Card Approval Prediction System that analyzes applicant details (income, credit score, employment, age, etc.) and predicts the likelihood of approval. • Provides fast, accurate and consistent decisions • Reduces fraud and default risk • Automates the approval process and saves time • Improves customer experience and satisfaction • Helps banks make data-driven credit decisions	
4. EMOTIONS BEFORE / AFTER [EM] BEFORE • Uncertainty • Delay • Frustration • Anxiety about approval AFTER • Quick approval • Transparency • Satisfaction • Confidence & trust	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Bank Websites • Mobile Banking Apps • Credit Card Application Portals • Email / SMS Notifications OFFLINE • Bank Branches • Customer Service Centers • Call Centers • Printed Application Forms	